CT BuildFest 2.0 – Problem Statements

30-Hour Coding Hackathon · 30–31 October 2026 ·

CT Group of Institutions, Maqsudan, Jalandhar

About this document

CT BuildFest 2.0 offers 48 problem statements across five themes: 12 each in Theme 1 (AI & Intelligent Systems) and Theme 3 (Smart Education & Social Innovation), and eight each in Themes 2, 4 and 5. The 30-hour hackathon runs on 30–31 October 2026 at CT Group of Institutions, Maqsudan, Jalandhar.

Teams of 3–6 pick one statement. Teams may also propose their own problem under theme (Student Innovation), provided they submit it in the same format at registration.

Organizer: CT Group of Institutions, powered by QuickHeal Academy, in association with ISTE. Industry Technology Session Partner: IBM, Knowledge Partner: upGrad

Theme 1 · AI & Intelligent Systems

Department for all statements: Computer Science & Engineering / AI-ML. Organization: CT Group of Institutions, Jalandhar.

PS ID	Title	Category	Theme
CTBF-AI-01	Multilingual crop-disease and farm advisory assistant	Software	Agriculture, FoodTech & Rural Development
CTBF-AI-02	AI triage assistant for rural health centres	Software	MedTech / HealthTech
CTBF-AI-03	Deepfake and fake-news detector for regional-language media	Software	Digital Media & Misinformation
CTBF-AI-04	Adaptive traffic control/Management from CCTV feeds	Software	Transportation & Smart Cities
CTBF-AI-05	Resume–job matching and skill-gap coach for tier-2 graduates	Software	Skill Development & Employability
CTBF-AI-06	Predictive maintenance for small manufacturing units	Software	Industry 4.0 & MSME Manufacturing
CTBF-AI-07	Government scheme eligibility navigator chatbot	Software	E-Governance & Citizen Services
CTBF-AI-08	Student wellbeing check-in with counsellor escalation	Software	Mental Health & Student Wellbeing
CTBF -AI-09	Plain-Language Legal Document and FIR Explainer	Software	Legal Aid & Citizen Services
CTBF -AI-10	Rural Power-Outage and Transformer-Fault Predictor	Software	Energy & Utilities
CTBF -AI-11	Task-Scam and Fake Job-Offer Detector	Software	Consumer Protection & Public Safety
CTBF -AI-12	Flood Early-Warning and Village Evacuation Advisor	Software	Disaster Management & Climate Resilience

CTBF-AI-01 · Multilingual Crop-Disease and Farm Advisory Assistant

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Agriculture, FoodTech & Rural Development

Background. Punjab farmers lose yield every season to late-detected crop diseases and pests. Extension officers are few, and most advisory apps are English-only.

Description. Build an assistant where a farmer photographs a leaf or asks by voice in Punjabi, Hindi or English and gets an answer in the same language.

- Detect at least 5 diseases or pests of wheat, paddy or a local vegetable, with a confidence score.
- Recommend treatment: product, dose per acre, timing and safety precautions.
- Below a confidence threshold, refuse to guess and refer to the nearest Krishi Vigyan Kendra.
- Run on a low-end Android phone over a weak 2G/3G connection.

Expected Solution. Working app showing image diagnosis and a voice Q&A loop in two languages, tested live on sample leaves.

CTBF-AI-02 · AI Triage Assistant for Rural Health Centres

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** MedTech / HealthTech

Background. Primary Health Centres see long queues with one doctor and limited staff. Critical patients often wait behind routine cases.

Description. Build a triage tool a nurse or ASHA worker uses at registration.

- Capture symptoms, vitals (BP, SpO2, temperature, pulse) and history through a simple form or voice.
- Assign a priority level (Emergency, Urgent, Routine) with the reasons shown.
- Flag red-flag combinations (e.g. chest pain with low SpO2) for immediate attention.
- Store records locally and sync when connectivity returns; the tool assists and never replaces the doctor.

Expected Solution. Working triage app with a live queue screen sorted by priority, demonstrated on judge-supplied patient cases.

CTBF-AI-03 · Deepfake and Fake-News Detector for Regional-Language Media

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Digital Media & Misinformation

Background. Manipulated videos, cloned voices and false forwards in Punjabi and Hindi spread faster than fact-checks, especially on WhatsApp.

Description. Build a verification tool a citizen or journalist can use before forwarding content.

- Accept text, image, audio or short video.
- Score the likelihood of manipulation (face swap, voice clone, edited image) with highlighted evidence.
- For text claims, search trusted sources and show matching fact-checks.
- Explain the verdict in plain Punjabi, Hindi or English.

Expected Solution. Web app or WhatsApp-style bot that correctly flags most items in a judge-supplied set of real and manipulated media.

CTBF-AI-04 · Adaptive Traffic Control and Management from CCTV Feeds

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Transportation & Smart Cities

Background. Fixed-timer signals at busy Jalandhar junctions cause long waits on one road while another stands empty. Ambulances get stuck in the queue, and congestion, wrong-side driving, illegal roadside parking and stalled vehicles go unnoticed until traffic police arrive.

Description. Build a system that reads live or recorded junction and road CCTV video to control signal timing and help the traffic control room manage the wider corridor.

- Detect and count cars, two-wheelers, autos, buses and trucks per lane.
- Signal control: compute green-time per approach from queue length and wait time, and coordinate adjacent junctions to form a "green wave".
- Give priority to detected emergency vehicles and create a green corridor along their route.
- Traffic management: detect congestion build-up, stalled or broken-down vehicles, wrong-side driving, red-light jumping and illegal parking, and alert the control room with a snapshot and location.
- Show a control-room dashboard with live junction status, an incident log and average wait time versus a fixed-timer baseline.

Expected Solution. Simulation (e.g. SUMO) driven by real video detections, showing a measurable reduction in average wait time, plus at least two traffic incidents (e.g. wrong-side driving, stalled vehicle) detected and alerted from sample footage.

CTBF-AI-05 · Resume–Job Matching and Skill-Gap Coach for Tier-2 Graduates

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Skill Development & Employability

Background. Many graduates from tier-2 and tier-3 colleges apply widely but are rejected by automated screening without knowing why.

Description. Build a career assistant for final-year students.

- Parse a resume (PDF/DOCX) and a job description.
- Give a match score with missing skills and keywords listed.
- Suggest free courses and a 30-day project plan to close each gap.
- Run a short AI mock interview for the role and score the answers.

Expected Solution. Working web app tested on real student resumes and live job postings, with a clear before/after match score.

CTBF-AI-06 · Predictive Maintenance for Small Manufacturing Units

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Industry 4.0 & MSME Manufacturing

Background. Jalandhar's sports-goods, hand-tool and auto-part units run machines until they fail. A single breakdown can halt an order.

Description. Build a low-cost predictive maintenance system for an MSME shop floor.

- Ingest vibration, temperature and current readings (real sensors or a supplied dataset).
- Detect anomalies and estimate remaining useful life per machine.
- Send alerts by SMS or WhatsApp to the supervisor in plain language.
- Keep cost low enough for a small unit: phone or Raspberry Pi as the hub.

Expected Solution. Dashboard with live machine health and a demonstrated early warning on a simulated fault.

CTBF-AI-07 · Government Scheme Eligibility Navigator Chatbot

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** E-Governance & Citizen Services

Background. Central and Punjab government schemes for students, farmers, women and senior citizens go unclaimed because people do not know they qualify.

Description. Build a conversational assistant that matches a citizen to schemes.

- Ask simple questions (age, income, occupation, category, district) in Punjabi, Hindi or English.
- Return eligible schemes with benefits, documents needed and the official application link.
- Answer follow-up questions grounded only in official scheme documents, with the source shown.
- Store no personal data without consent.

Expected Solution. Working chatbot covering at least 20 real schemes, answering judge questions accurately with citations.

CTBF-AI-08 · Student Wellbeing Check-In with Counsellor Escalation

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Mental Health & Student Wellbeing

Background. Exam stress, homesickness and isolation among hostel students often go unnoticed until they become serious. Counsellors cannot check on everyone.

Description. Build a private, opt-in wellbeing companion for college students.

- Short daily or weekly mood check-ins by text or voice.
- Detect sustained low mood or risk language and gently suggest talking to a counsellor.
- Escalate high-risk signals to the campus counsellor only under a consented, clearly explained protocol; show helpline numbers such as Tele-MANAS (14416).
- Give counsellors anonymized, aggregate trends by hostel or year.

Expected Solution. Working app with a student view and a counsellor dashboard, plus a written privacy and safety policy.

CTBF-AI-09 · Plain-Language Legal Document and FIR Explainer

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Legal Aid & Citizen Services

Background. Citizens sign rent agreements, loan papers and land documents, or receive court notices and FIR copies, without understanding them, and legal advice is costly.

Description. Build an AI tool that explains legal documents clause by clause in plain Punjabi, Hindi or English, flags risky clauses and deadlines, and points users to free legal aid without storing their documents or giving legal advice

- Upload a document and get a clause-by-clause summary in plain Punjabi, Hindi or English.
- Highlight risky clauses (penalties, auto-renewal, waived rights) and important deadlines.
- Explain what the person can do next and link to free legal aid through DLSA or NALSA.
- Give information only, not legal advice, and store no documents without consent.

Expected Solution. A working tool that explains 5 judge-supplied documents accurately and correctly spots planted risky clauses.

CTBF-AI-10 · Rural Power-Outage and Transformer-Fault Predictor

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Energy & Utilities

Background. Transformer failures in rural feeders, especially during paddy season, leave villages and tubewells without power for days, and repair crews react only after a failure.

Description. Build an AI system that predicts which rural transformers are at risk of failing from load, weather and past fault data, ranks them for preventive maintenance, and groups outage reports sent by missed call or SMS on a live dashboard for the utility.

- Use load, temperature, weather and past fault data (real or synthetic) to predict which transformers are at risk of failing.
- Rank transformers for preventive maintenance and suggest load-balancing actions.
- Let consumers report outages by missed call or SMS and group the reports to locate faults.
- Show the utility a dashboard of predicted risk, live outages and restoration times.

Expected Solution. A working dashboard on a sample feeder dataset that shows early warnings for simulated faults and outage reports grouped onto a map.

CTBF-AI-11 · Task-Scam and Fake Job-Offer Detector

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Consumer Protection & Public Safety

Background. Students and homemakers are drawn in by “like videos and earn” Telegram tasks, fake work-from-home jobs and deposit-based job offers, then lose money in stages.

Description. Build an AI tool that detects task scams and fake job offers in posts, chats and offer letters, verifies the company against MCA records, and warns users before their next payment, with one-tap reporting to helpline 1930.

- Analyse a job post, Telegram or WhatsApp chat or offer letter for scam patterns: upfront deposits, task-to-prepaid escalation, unverifiable companies, pressure to move to private chats.
- Check the company against the MCA company registry and its official domain.
- Identify which stage of a known scam script the victim is at and warn them before the next payment.
- Explain the risk in simple language and give one-tap reporting to helpline 1930.

Expected Solution. A working bot or web app that correctly classifies a judge-supplied set of genuine and scam job offers and chat logs.

CTBF-AI-12 · Flood Early-Warning and Village Evacuation Advisor

Track: AI & Intelligent Systems · **Category:** Software · **Theme:** Disaster Management & Climate Resilience

Background. Recent floods along the Sutlej, Beas and Ghaggar hit villages with little warning, and evacuation messages arrived late or were too vague.

Description. Build an AI system that predicts village-level flood risk 24–72 hours ahead using rainfall, river and dam data, maps safe evacuation routes, and sends Punjabi voice or SMS alerts, with residents’ photo reports keeping the map up to date.

- Combine rainfall forecasts, river gauge and dam-release data, and elevation maps to predict flood risk per village 24–72 hours ahead.
- Show a risk map with safe evacuation routes and the nearest relief camps.
- Send voice or SMS alerts in Punjabi tailored to each village’s risk level.
- Let residents report waterlogging or embankment breaches with photos to update the map.

Expected Solution. A working dashboard that replays a past flood event on sample data and shows which villages would have been warned, and how early.

Theme 2 · Cyber Security

Department for all statements: Cyber Security & Digital Safety Cell, with QuickHeal Academy. Organization: CT Group of Institutions, Jalandhar. All solutions must be defensive; testing only on systems the team owns or is permitted to test.

PS ID	Title	Category	Theme
CTBF-CS-01	Real-time scam shield for UPI and payment frauds	Software	FinTech & Digital Payment Security
CTBF-CS-02	Voice-clone and “digital arrest” call detector	Software	Telecom Fraud & Public Safety
CTBF-CS-03	Lightweight intrusion detection for campus networks	Software	Network Security
CTBF-CS-04	Tamper-proof academic certificate verification	Software	Blockchain & Document Security
CTBF-CS-05	Ransomware early-warning agent for small businesses	Software	Endpoint Security for MSMEs
CTBF-CS-06	Android app privacy auditor	Software	Mobile Security & Data Privacy
CTBF-CS-07	Verifiable e-voting for student council elections	Software	Blockchain & E-Governance
CTBF-CS-08	Home IoT device security checker	Software + Hardware	IoT Security

CTBF-CS-01 · Real-Time Scam Shield for UPI and Payment Frauds

Track: Cyber Security · **Category:** Software · **Theme:** FinTech & Digital Payment Security

Background. Fake collect requests, phishing links, fake customer-care numbers and malicious QR codes drain money from students, seniors and first-time digital users.

Description. Build a tool that warns a user before they lose money.

- Classify a pasted SMS or WhatsApp message as Safe, Suspicious or Scam, with a plain-language reason.
- Check URLs for phishing signs: look-alike domains, newly registered domains, shorteners.
- Decode a QR code and warn if it is a payment request rather than a receive request.
- Explain warnings in English, Hindi and Punjabi; analyse messages on-device or anonymised.

Expected Solution. App, browser extension or web app that correctly classifies a judge-supplied set of real and scam messages, links and QR codes, with one-tap reporting to cybercrime.gov.in or helpline 1930.

CTBF-CS-02 · Voice-Clone and "Digital Arrest" Call Detector

Track: Cyber Security · **Category:** Software · **Theme:** Telecom Fraud & Public Safety

Background. Fraudsters pose as police, CBI or a relative in distress, sometimes with AI-cloned voices, and pressure victims into transfers.

Description. Build a call-safety assistant.

- Analyse a recorded or live call audio and estimate whether the voice is synthetic.
- Transcribe and flag pressure tactics: arrest threats, secrecy demands, urgent transfer requests.
- Show an on-screen warning and a "verify with family" prompt during the call.
- Process audio on-device or delete it after analysis.

Expected Solution. Working prototype that flags synthetic voices and scam scripts in a judge-supplied set of audio clips.

CTBF-CS-03 · Lightweight Intrusion Detection for Campus Networks

Track: Cyber Security · **Category:** Software · **Theme:** Network Security

Background. College networks carry thousands of student devices, many unpatched. Commercial SIEM tools are too costly for most institutions.

Description. Build a low-cost network monitoring and intrusion detection system.

- Ingest network logs or packet captures (live lab network or datasets such as CIC-IDS).
- Detect port scans, brute-force logins, malware beaconing and unusual data transfers.
- Rank alerts by severity with a suggested response for the network admin.
- Run on a single low-cost machine or Raspberry Pi.

Expected Solution. Dashboard that detects attacks replayed from a test capture, with a low false-positive rate on normal traffic.

CTBF-CS-04 · Tamper-Proof Academic Certificate Verification

Track: Cyber Security · **Category:** Software · **Theme:** Blockchain & Document Security

Background. Forged degrees and mark sheets slip through hiring and admissions because manual verification takes weeks.

Description. Build a system for colleges to issue and employers to verify certificates instantly.

- Issue certificates with a digital signature and a secure QR code.
- Let anyone verify authenticity by scanning the QR or uploading the PDF.
- Detect edited fields (name, marks, dates) and show what was changed.
- Keep student data private: verification reveals only what the certificate states.

Expected Solution. Issuer portal and verifier app that catch a judge-tampered certificate during the demo.

CTBF-CS-05 · Ransomware Early-Warning Agent for Small Businesses

Track: Cyber Security · **Category:** Software · **Theme:** Endpoint Security for MSMEs

Background. Shops, clinics and CA offices rarely have IT staff. One ransomware infection can lock all their records.

Description. Build a lightweight defensive agent for Windows or Linux desktops.

- Monitor file activity for ransomware behaviour: mass renames, rapid encryption, entropy spikes, canary-file changes.
- Stop the offending process and alert the owner.
- Keep protected, versioned backups of key folders that the malicious process cannot reach.
- Use only safe simulators, never live ransomware, for testing.

Expected Solution. Agent that detects and halts a benign ransomware simulator within seconds and restores the affected files.

CTBF-CS-06 · Android App Privacy Auditor

Track: Cyber Security · **Category:** Software · **Theme:** Mobile Security & Data Privacy

Background. Many free apps request contacts, location, microphone and SMS access they do not need, and share data with third-party trackers.

Description. Build a tool that audits installed apps for users.

- List each app's permissions and flag those unusual for its category.
- Detect embedded trackers and ad SDKs.
- Give each app a privacy score with a plain-language explanation.
- Guide the user to revoke risky permissions in settings.

Expected Solution. Working Android app or APK-upload web tool that audits at least 10 popular apps live.

CTBF-CS-07 · Verifiable E-Voting for Student Council Elections

Track: Cyber Security · **Category:** Software · **Theme:** Blockchain & E-Governance

Background. Paper ballots for student elections are slow, and online forms are easy to manipulate and impossible to audit.

Description. Build a secure campus voting system.

- Authenticate each eligible student once (college ID + OTP).
- Keep the ballot secret while preventing double voting.
- Give each voter a receipt to confirm their vote was counted, without revealing the choice.
- Publish results that any observer can independently verify.

Expected Solution. Working system running a mock election among judges, with a published audit trail.

CTBF-CS-08 · Home IoT Device Security Checker

Track: Cyber Security · **Category:** Software + Hardware · **Theme:** IoT Security

Background. Wi-Fi cameras, smart plugs and routers often ship with default passwords and open ports, making homes easy targets.

Description. Build a scanner for a user's own home network.

- Discover all devices on the network and identify their type.
- Check for default credentials, open risky ports and outdated firmware.
- Show a home security score with step-by-step fixes.
- Scan only the network the user owns, with clear consent.

Expected Solution. Working tool (Raspberry Pi or phone app) that audits a demo network with at least 3 IoT devices and reports fixable issues.

Theme 3 · Smart Education & Social Innovation

Department for all statements: Education Technology & Social Outreach Cell. Organization: CT Group of Institutions, Jalandhar.

PS ID	Title	Category	Theme
CTBF-SE-01	Offline adaptive learning app for government-school students	Software	EdTech & Personalised Learning
CTBF-SE-02	School/College dropout early-warning system	Software	Education Analytics
CTBF-SE-03	Indian Sign Language to text and speech translator	Software	Accessibility & Inclusion
CTBF-SE-04	Virtual science lab for schools/College without labs	Software	EdTech & Virtual Labs
CTBF-SE-05	Emergency blood-donor matching network	Software	MedTech / HealthTech (Emergency Care)
CTBF-SE-06	Surplus food rescue platform	Software	Food Security & Hunger Relief
CTBF-SE-07	Apprenticeship marketplace for ITI and diploma students	Software	Skill Development & Employability
CTBF-SE-08	Indoor campus navigation for visually impaired students	Software + Hardware	Accessibility & Inclusion
CTBF-SE-09	Multilingual Career Counselling Kiosk for Walk-in Visitors	Software	Skill Development & Employability
CTBF-SE-10	Hands-on Project Evaluator with Rubric Feedback	Software	EdTech & Assessment
CTBF-SE-11	Spoken-English and Presentation Coach for Vernacular Learners	Software	Skill Development & Employability
CTBF-SE-12	WhatsApp Learner Helpdesk Agent	Software	Learner Services & Operations

CTBF-SE-01 · Offline Adaptive Learning App for Government-School Students

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** EdTech & Personalised Learning

Background. Students in rural government schools learn at very different levels within one class, and many lack steady internet at home.

Description. Build a learning app that adapts to each student's level.

- Short diagnostic test to place the student on a skill map (maths and science, classes 6–10).
- Personalised practice that gets harder or easier based on answers.
- Works fully offline and syncs progress when connected; Punjabi, Hindi and English.
- Teacher view showing which students are stuck on which concept.

Expected Solution. Working app with at least one complete chapter mapped, demonstrating adaptation across two simulated students.

CTBF-SE-02 · School/College Dropout Early-Warning System

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** Education Analytics

Background. Students in schools and colleges rarely drop out suddenly. Falling attendance, marks, backlogs and fee delays show the risk months earlier, but nobody connects the signals.

Description. Build a system that flags at-risk students early in both schools and colleges.

- Combine attendance, marks, fee status and teacher remarks; for colleges also add semester results, backlogs/re-appears, hostel status and mentor notes.
- Predict dropout risk per student with the main contributing factors.
- Suggest interventions: parent call, mentor meeting, scholarship link, remedial class or counselling.
- Track whether each intervention improved the student's indicators.
- Support separate configurations for a school (class/section) and a college (department/semester).

Expected Solution. Dashboard on a synthetic or anonymised dataset of at least 500 students covering both school and college records, with explainable risk scores.

CTBF-SE-03 · Indian Sign Language to Text and Speech Translator

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** Accessibility & Inclusion

Background. Deaf and hard-of-hearing students struggle to communicate with teachers and classmates who do not know Indian Sign Language (ISL).

Description. Build a real-time ISL translator.

- Recognise at least 30 common ISL signs (alphabets, numbers, classroom words) from a webcam or phone camera.
- Convert them to text and speech in English or Hindi.
- Reverse mode: convert spoken or typed text to sign animations or images.
- Run in real time on a laptop or mid-range phone.

Expected Solution. Live demo translating signs performed by the team, with measured accuracy.

CTBF-SE-04 · Virtual Science Lab for Schools/College Without Labs

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** EdTech & Virtual Labs

Background. Many schools and colleges lack equipment for physics, chemistry and basic engineering practicals, so students learn experiments only from textbooks.

Description. Build an interactive virtual lab.

- Simulate at least 5 experiments: at least 3 from the school board syllabus (e.g. Ohm's law, pendulum, titration, refraction) and at least 2 from first-year college labs (e.g. Wheatstone bridge, diode V-I characteristics, Young's modulus).
- Let students change variables, take readings and plot graphs.
- Include a viva quiz and auto-generated lab record, with a School or College mode matched to the syllabus level.
- Run in a browser on low-end devices.

Expected Solution. Working web lab with 5 experiments (school and college level) that a judge can perform end to end.

CTBF-SE-05 · Emergency Blood-Donor Matching Network

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** MedTech / HealthTech (Emergency Care)

Background. Families search WhatsApp groups for rare blood groups in emergencies, while willing donors nearby never hear about the request.

Description. Build a donor-matching platform.

- Register donors with blood group, location and last donation date (eligibility auto-checked).
- Broadcast urgent requests to eligible donors within a radius, nearest first.
- Let blood banks update live stock by group.
- Protect donor phone numbers until the donor accepts.

Expected Solution. Working app showing a request matched to donors and a blood bank in a live demo.

CTBF-SE-06 · Surplus Food Rescue Platform

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** Food Security & Hunger Relief

Background. Hostels, canteens, weddings and restaurants discard cooked food daily while nearby shelters and NGOs run short.

Description. Build a platform that redirects surplus food safely.

- Donors post quantity, food type, pickup window and preparation time.
- Match to the nearest NGO or volunteer with capacity; route the pickup.
- Enforce food-safety rules (time since cooking, storage).
- Track meals saved and show impact to donors.

Expected Solution. Working donor, NGO and volunteer apps demonstrating a complete pickup flow.

CTBF-SE-07 · Apprenticeship Marketplace for ITI and Diploma Students

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** Skill Development & Employability

Background. Local MSMEs need trained hands, and ITI students need apprenticeships, but there is no simple local channel connecting them.

Description. Build a local skills marketplace.

- Students build a skills profile with short video proof of work.
- Employers post apprenticeships and shortlist by skill, trade and distance.
- Track attendance and supervisor feedback during the apprenticeship.
- Issue a verified digital completion certificate.

Expected Solution. Working platform with sample employers and students demonstrating posting, matching and certification.

CTBF-SE-08 · Indoor Campus Navigation for Visually Impaired Students

Track: Smart Education & Social Innovation · **Category:** Software + Hardware · **Theme:** Accessibility & Inclusion

Background. GPS does not work well indoors, so visually impaired students depend on others to find classrooms, labs and washrooms.

Description. Build an indoor navigation aid.

- Locate the user indoors using BLE beacons, QR markers or computer vision.
- Give turn-by-turn voice directions to a chosen room.
- Warn of obstacles such as stairs and doors.
- Work hands-free with voice commands.

Expected Solution. Working prototype guiding a blindfolded user between two rooms on the hackathon floor.

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CTBF-SE-09 · Multilingual Career Counselling Kiosk for Walk-in Visitors

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** Skill Development & Employability

Background. People who walk into upskilling centres in tier-2 cities are often unsure which course suits their background, and counsellors are stretched thin during peak hours.

Description. Build a self-service career counselling kiosk.

- Ask about education, experience, interests and goals in Punjabi, Hindi or English.
- Recommend 2–3 learning pathways with duration, prerequisites, likely roles and realistic salary ranges from public data.
- Point visitors who lack the prerequisites to free foundation resources.
- Book a human counsellor slot and hand over a summary of the visitor's profile.

Expected Solution. Working kiosk app giving clear, justified recommendations for 5 judge-supplied visitor profiles.

CTBF-SE-10 · Hands-on Project Evaluator with Rubric Feedback

Track: Smart Education & Social Innovation · **Category:** Software · **Theme:** EdTech & Assessment

Background. Learners build projects for their portfolios, but manual review takes days, and the feedback is often generic.

Description. Build an automated project evaluation tool.

- Accept a GitHub repo or notebook (Python, SQL or web app) along with the project brief.
- Check that it runs, along with test coverage and code quality, and score it against a mentor-defined rubric.
- Give line-level, actionable feedback and a portfolio-readiness score.
- Flag possible plagiarism and check the learner's AI-use disclosure.

Expected Solution. Working evaluator scoring 5 judge-supplied projects, with feedback a mentor reviews and agrees with.

CTBF-SE-11 · Spoken-English and Presentation Coach for Vernacular Learners

Track: Smart Education & Social Innovation · **Category:** Software + Hardware · **Theme:** Skill Development & Employability

Background. Many technically strong students and job-seekers in Punjab lose interviews because they lack confidence in spoken English, and private coaching is expensive.

Description. Build an AI speaking coach.

- Score fluency, pace, filler words, pronunciation and grammar as the learner speaks on a prompt or presents a slide.
- Explain each mistake in Punjabi or Hindi alongside the corrected English sentence.
- Offer role-plays (self-introduction, project walkthrough, client call) whose difficulty adapts to the learner.
- Track progress over weeks and run on a mid-range phone.

Expected Solution. Working app assessing live speech from 2 team members, with clear feedback and a progress view.

CTBF-SE-12 · WhatsApp Learner Helpdesk Agent

Track: Smart Education & Social Innovation · **Category:** Software + Hardware · **Theme:** Learner Services & Operations

Background. Staff at colleges and training centres spend hours answering the same questions about timetables, fees, certificates and makeup classes, often after hours.

Description. Build a WhatsApp-style AI helpdesk.

- Answer in English, Hindi or Punjabi using only the institution's own policy documents, with the source shown.
- Complete simple actions such as rescheduling a class, requesting a certificate or booking a counsellor, using a mock student database.
- Say it doesn't know rather than guessing, and hand over to a human with the full conversation.
- Show staff the most common unresolved queries each week.

Expected Solution. Working bot answering 15 judge questions accurately with citations and completing 2 actions end to end.

Theme 4 · IoT, Robotics & Automation

Department for all statements: Electronics & Communication / Mechanical Engineering. Organization: CT Group of Institutions, Jalandhar. Teams bring their own components; a basic kit list should be shared at registration.

PS ID	Title	Category	Theme
CTBF-IR-01	Smart irrigation controller to save groundwater	Hardware + Software	Agriculture & Water Conservation
CTBF-IR-02	Low-cost air-quality network with stubble-fire alerts	Hardware + Software	Air Pollution & Environmental Monitoring
CTBF-IR-03	Automatic waste-segregation bin	Hardware + Software	Waste Management
CTBF-IR-04	Campus energy monitoring and auto-cutoff	Hardware + Software	Energy Efficiency & Smart Buildings
CTBF-IR-05	Sewer and manhole inspection robot	Hardware + Software	Robotics for Sanitation & Worker Safety
CTBF-IR-06	Two-wheeler crash detection and automatic SOS	Hardware + Software	Smart Vehicles & Road Safety
CTBF-IR-07	Cold-chain monitor for milk and vaccines	Hardware + Software	Supply Chain & Cold-Chain Logistics
CTBF-IR-08	Fall-detection wearable for elderly people	Hardware + Software	HealthTech & Elderly Care

CTBF-IR-01 · Smart Irrigation Controller to Save Groundwater

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Agriculture & Water Conservation

Background. Punjab's groundwater is falling fast, and flood irrigation wastes much of what is pumped.

Description. Build an automatic irrigation controller for a small field or nursery.

- Read soil moisture at two or more depths plus temperature and humidity.
- Switch the pump or valve on and off automatically, skipping irrigation when rain is forecast.
- Show water used and saved on a phone app; control the pump remotely.
- Run on solar or battery with a low-cost microcontroller (ESP32 or similar).

Expected Solution. Working prototype watering a soil tray automatically, with live data on a dashboard.

CTBF-IR-02 · Low-Cost Air-Quality Network with Stubble-Fire Alerts

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Air Pollution & Environmental Monitoring

Background. Every October and November, stubble burning and traffic push air quality in Punjab to hazardous levels. Official monitors are few and far apart.

Description. Build a network of low-cost air-quality nodes.

- Measure PM2.5, PM10, CO and temperature/humidity per node.
- Calibrate readings against a reference and show a live AQI map.
- Detect sudden smoke spikes and alert residents and authorities.
- Send health advice by AQI level (mask, avoid outdoor exercise).

Expected Solution. At least two working nodes reporting to a live map, with a demonstrated smoke alert.

CTBF-IR-03 · Automatic Waste-Segregation Bin

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Waste Management

Background. Mixed waste at source makes recycling hard, and manual sorting is unsafe for workers.

Description. Build a bin that sorts waste as it is dropped in.

- Classify items as wet, dry/recyclable, metal or hazardous using sensors and/or a camera.
- Route each item to the right compartment mechanically.
- Report fill levels and request collection when full.
- Keep the build cost under a stated budget (teams to declare).

Expected Solution. Working bin correctly sorting a judge-supplied mix of 10 items, with fill levels and sorting counts shown on a web or mobile dashboard.

CTBF-IR-04 · Campus Energy Monitoring and Auto-Cutoff

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Energy Efficiency & Smart Buildings

Background. Fans, lights and ACs run in empty classrooms and labs, inflating the campus power bill.

Description. Build a smart energy system for classrooms.

- Detect occupancy with PIR or camera-based counting.
- Switch off loads after a set time of no occupancy; allow manual override.
- Meter energy per room and show a ranking dashboard.
- Alert facility staff to abnormal consumption.

Expected Solution. Working prototype controlling at least two loads, with energy saved shown live.

CTBF-IR-05 · Sewer and Manhole Inspection Robot

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Robotics for Sanitation & Worker Safety

Background. Workers still enter sewers for inspection and cleaning, risking death from toxic gases. Manual scavenging is banned, yet the need for inspection remains.

Description. Build a remotely operated inspection robot.

- Enter a pipe or manhole model and stream live video with lighting.
- Measure toxic gases (H2S, methane) and send readings to the operator.
- Map blockage locations along the pipe.
- Be waterproof enough for a wet demo environment.

Expected Solution. Working robot navigating a mock pipe, streaming video and gas readings to an operator app or dashboard that marks blockage points.

CTBF-IR-06 · Two-Wheeler Crash Detection and Automatic SOS

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Smart Vehicles & Road Safety

Background. In two-wheeler accidents on highways and rural roads, injured riders often lie unattended because nobody calls for help.

Description. Build a helmet- or bike-mounted crash detector.

- Detect a crash using an accelerometer and gyroscope, separating it from potholes and hard braking.
- Give a 30-second cancel window, then send location to family and dial 112.
- Detect helmet worn and alcohol (optional) before ignition.
- Run on a small rechargeable battery.

Expected Solution. Working prototype triggering an SOS with live location on a simulated crash.

CTBF-IR-07 · Cold-Chain Monitor for Milk and Vaccines

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** Supply Chain & Cold-Chain Logistics

Background. Milk collection centres and rural vaccine stores lose stock when refrigeration fails unnoticed during power cuts.

Description. Build a cold-chain monitoring device.

- Log temperature and humidity continuously with a battery backup.
- Alert by SMS when limits are crossed, even without Wi-Fi (GSM fallback).
- Keep a tamper-proof log for audits.
- Predict time-to-spoil after a cooling failure.

Expected Solution. Working logger in a cooler box that alerts within a minute of a simulated failure.

CTBF-IR-08 · Fall-Detection Wearable for Elderly People

Track: IoT, Robotics & Automation · **Category:** Hardware + Software · **Theme:** HealthTech & Elderly Care

Background. Many elderly people in Punjab live alone while children work abroad or in cities. A fall can go unnoticed for hours.

Description. Build a wearable that detects falls and calls for help.

- Detect falls with motion sensors, separating them from sitting or lying down.
- Alert family and a neighbour with location if the user does not respond.
- Include an SOS button and optional heart-rate monitoring.
- Last at least 24 hours on one charge.

Expected Solution. Working wearable detecting simulated falls with few false alarms, plus a caregiver app.

Theme 5 · Sustainability, Environment & Student Innovation

Department for all statements: Civil & Environmental Engineering / Innovation Cell. Organization: CT Group of Institutions, Jalandhar.

PS ID	Title	Category	Theme
CTBF-SU-01	Paddy-stubble marketplace linking farmers to biomass buyers	Software	Agriculture & Crop Residue Management
CTBF-SU-02	Campus carbon-footprint tracker	Software	Clean & Green Technology (Climate Action)
CTBF-SU-03	Village groundwater budgeting dashboard	Software + Hardware	Water Resource Management
CTBF-SU-04	Industrial effluent monitoring for drains and rivers	Hardware	Water Pollution Control
CTBF-SU-05	E-waste collection and refurbish-reuse platform	Software	E-Waste & Circular Economy
CTBF-SU-06	Urban heat and tree-plantation planner using satellite data	Software	Climate Resilience & Urban Planning
CTBF-SU-07	Canteen and mess food-waste reduction system	Software + Hardware	Food Waste Management
CTBF-SU-08	Student Innovation – open problem statement	Software / Hardware	Open Innovation (team-defined)

CTBF-SU-01 · Paddy-Stubble Marketplace Linking Farmers to Biomass Buyers

Track: Sustainability, Environment & Student Innovation · **Category:** Software · **Theme:** Agriculture & Crop Residue Management

Background. Farmers burn paddy stubble because the window before wheat sowing is short and collection is not arranged in time. Biomass power plants, CBG plants and pellet makers need that same straw.

Description. Build a marketplace for crop residue.

- Farmers list acreage, location and harvest date by voice in Punjabi.
- Match listings to buyers and baler/machinery owners, and schedule pickups by route.
- Show transparent pricing and track payments.
- Report straw diverted from burning by village.

Expected Solution. Working app with farmer, buyer and machinery-owner flows on sample Jalandhar-district data.

CTBF-SU-02 · Campus Carbon-Footprint Tracker

Track: Sustainability, Environment & Student Innovation · **Category:** Software · **Theme:** Clean & Green Technology (Climate Action)

Background. Colleges set green goals but cannot measure emissions from electricity, diesel generators, transport and waste.

Description. Build a carbon accounting tool for an institution.

- Import bills and logs (electricity, diesel, LPG, bus routes, waste).
- Compute emissions by source using standard emission factors.
- Suggest reductions with estimated savings (solar, EV buses, LED retrofit).
- Let students log personal commute and earn green points.

Expected Solution. Dashboard computing a full campus footprint from sample data, with a ranked reduction plan.

CTBF-SU-03 · Village Groundwater Budgeting Dashboard

Track: Sustainability, Environment & Student Innovation · **Category:** Software + Hardware · **Theme:** Water Resource Management

Background. Most Punjab blocks are over-exploited for groundwater, yet villages have no simple view of how much water they extract versus recharge.

Description. Build a water-budget tool for a village or block.

- Collect well-level readings via a low-cost sensor or farmer SMS reports.
- Estimate recharge from rainfall and extraction from crop area and pump hours.
- Show the water balance and suggest crop shifts (e.g. maize, pulses) or recharge pits.
- Present it simply for the gram panchayat.

Expected Solution. Working dashboard on sample data, plus one sensor node or SMS-reporting flow.

CTBF-SU-04 · Industrial Effluent Monitoring for Drains and Rivers

Track: Sustainability, Environment & Student Innovation · **Category:** Hardware · **Theme:** Water Pollution Control

Background. Untreated effluent from leather, electroplating and dyeing units reaches drains that feed Punjab's rivers. Violations are found late, if at all.

Description. Build a real-time water-quality monitoring node.

- Measure pH, TDS, turbidity and temperature continuously.
- Detect sudden discharge events and timestamp them.
- Upload data to a public dashboard and alert the pollution control board.
- Run on solar power at a drain outlet.

Expected Solution. Working node detecting a simulated discharge in a water sample, with a live dashboard.

CTBF-SU-05 · E-Waste Collection and Refurbish-Reuse Platform

Track: Sustainability, Environment & Student Innovation · **Category:** Software · **Theme:** E-Waste & Circular Economy

Background. Old phones and laptops sit in drawers or go to informal recyclers who burn wires and leak toxins.

Description. Build a platform for responsible e-waste handling.

- Let users list devices and get an estimated value (refurbish, parts, recycle).
- Schedule pickup with authorised recyclers or refurbishers.
- Wipe-data checklist and certificate for the user.
- Route refurbished devices to students who need them.

Expected Solution. Working platform showing listing, pickup scheduling and donation matching.

CTBF-SU-06 · Urban Heat and Tree-Plantation Planner Using Satellite Data

Track: Sustainability, Environment & Student Innovation · **Category:** Software · **Theme:** Climate Resilience & Urban Planning

Background. Summer heat in Punjab's cities keeps rising, and plantation drives often plant in the wrong places.

Description. Build a planning tool for a city ward.

- Map land-surface temperature and green cover from free satellite data (Landsat/Sentinel).
- Identify the hottest, least-green zones.
- Recommend plantation spots and native species.
- Track planted trees with geotagged photos and survival rates.

Expected Solution. Interactive map of Jalandhar with heat hotspots and a ranked plantation plan.

CTBF-SU-07 · Canteen and Mess Food-Waste Reduction System

Track: Sustainability, Environment & Student Innovation · **Category:** Software + Hardware · **Theme:** Food Waste Management

Background. Hostel messes cook to fixed quantities, and much is thrown away daily.

Description. Build a system to cut mess food waste.

- Students confirm meals in advance; the system forecasts demand per dish.
- Weigh leftover waste per meal with a smart scale.
- Show waste trends and dishes most often wasted.
- Link surplus to food-rescue NGOs (see CTBF-SE-06).

Expected Solution. Working forecast app and a weighing prototype, with a waste-trend dashboard.

CTBF-SU-08 · Student Innovation – Open Problem Statement

Track: Sustainability, Environment & Student Innovation (Student Innovation) · **Category:** Software / Hardware · **Theme:** Open Innovation (team-defined)

Background. Some of the best ideas come from problems students see in their own villages, homes and campuses.

Description. Teams may propose their own problem under any of the five themes.

- Submit it in this same format: title, theme, category, background, description, expected solution.
- The problem must be real, specific and verifiable, with a clearly identified user.
- Organizers approve or refine proposals before the hackathon starts.

Expected Solution. A working prototype that solves the approved problem, judged on the same criteria as all other statements.

Evaluation and Submission

All 48 statements are judged on the same 100-point scale, so Software and Hardware entries compete fairly. The summary weights are below; the detailed rubric, scoring bands and judging process that follow explain exactly how each mark is awarded.

Criterion	Weight	What judges look for
Problem understanding	**15**	Clear user, real pain point, constraints respected
Innovation	**20**	Fresh approach, not a copy of an existing product
Technical implementation	**25**	Working prototype, sound architecture, code quality
Impact and feasibility	**20**	Scale of benefit, cost, ease of real-world rollout
Demo and presentation	**10**	Live demo works, clear pitch within time
Ethics, privacy and safety	**10**	Consent, data protection, responsible use

AI and Open-Source Attribution Policy

Teams may use open-source software, public datasets and AI tools, including Anthropic's Claude models, OpenAI's GPT models (for example ChatGPT or Codex) and any other AI model or assistant such as Gemini, Llama, GitHub Copilot or Hugging Face models. Using them is allowed; using them without saying so is not. Any team that takes help from open-source work, Anthropic models or GPT models must cite it in both its work (code repository and README) and its presentation deck.

What must be cited.

- **Open-source code and libraries:** project name, version, licence and link, for anything beyond standard language libraries; any code copied or adapted from GitHub, Stack Overflow or tutorials.
- **Pre-trained models and datasets:** model or dataset name, source, licence and link (e.g. YOLO, Whisper, PlantVillage, CIC-IDS).
- **AI models used inside the product:** provider and model (e.g. Anthropic Claude, OpenAI GPT), how it is accessed (API, app, local) and what it does in the solution.
- **AI tools used to build the project:** any AI assistant used to write or debug code, design the architecture, generate images, or draft slides, README or the pitch, with a one-line note on which parts it helped with.

Where to cite it.

- **README:** a section titled "AI and Open-Source Attribution" with the full list, in the format shown below.
- **Code:** a short comment at the top of any file or block that is substantially AI-generated or copied from an open-source source, e.g. "# Generated with Claude, reviewed and modified by the team".
- **PPT:** the tech-stack or architecture slide marks AI and open-source components, and the last slide, "References and AI / Open-Source Credits", lists them all.
- **Demo video:** a closing credits frame naming the main open-source projects and AI models used.

Citation format. One line per item: Name (provider/author), version or model, licence, link, how it was used.

Type	Example citation
Open-source library	Ultralytics YOLOv8, v8.2, AGPL-3.0, github.com/ultralytics/ultralytics: vehicle detection in CCTV frames.
Anthropic model	Claude (Anthropic), accessed via Claude API: classifies scam SMS in the app. Claude Code used to generate the Flask backend boilerplate; reviewed and edited by the team.
OpenAI GPT model	GPT model (OpenAI), via ChatGPT: drafted README text and slide 2 wording; reviewed and edited by the team.
Dataset	PlantVillage dataset, CC BY 4.0, link: training data for leaf-disease detection.